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Bayesian doubly adaptive elastic-net Lasso for VAR shrinkage

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ABSTRACT

We develop a novel Bayesian doubly adaptive elastic-net Lasso (DAELasso) approach for VAR shrinkage. DAELasso achieves variable selection and coefficient shrinkage in a data-based manner. It deals constructively with explanatory variables which tend to be highly collinear by encouraging the grouping effect. In addition, it also allows for different degrees of shrinkage for different coefficients. Rewriting the multivariate Laplace distribution as a scale mixture, we establish closed-form conditional posteriors that can be drawn from a Gibbs sampler. An empirical analysis shows that the forecast results produced by DAELasso and its variants are comparable to those from other popular Bayesian methods, which provides further evidence that the forecast performances of large and medium sized Bayesian VARs are relatively robust to prior choices, and, in practice, simple Minnesota types of priors can be more attractive than their complex and well-designed alternatives.
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1. Introduction

Tibshirani's (1996) least absolute shrinkage and selection operator (Lasso) and its variants, such as the elastic net Lasso (e-net Lasso) of Zou and Hastie (2005), and the grouped Lasso of Yuan and Lin (2007), are widely used for variable selection and parameter shrinkage for large data sets. Recently, the Bayesian Lasso has gained in popularity, as it can be implemented easily through MCMC or a Gibbs sampler (e.g., Kyung, Gill, Ghosh, & Casella, 2010; Park & Casella, 2008), and it can achieve adaptive shrinkage automatically, in order to allow for different degrees of shrinkage (e.g., Griffin & Brown, 2011; Leng, Tran, & Nott, 2010). Despite being successful, the Lasso literature has concentrated mainly on single equation models. To the best of our knowledge, only a few studies in the frequentist framework (e.g., Hsu, Hung, & Chang, 2008; Song & Bickel, 2011) have explored Lasso shrinkage for vector autoregressive (VAR) models; and these available methods can be too restrictive, as they either assume that the covariance matrix of the VAR errors is diagonal, or assume that its off-diagonal elements are much smaller than the diagonal ones.

This paper develops a novel Bayesian Lasso method for VAR shrinkage. Given that large VAR models usually have highly correlated explanatory variables, we propose using a doubly adaptive e-net Lasso (DAELasso) for macroeconomic research. DAELasso extends Zou and Zhang's (2009) adaptive e-net Lasso for single equation models into the VAR context. Unlike the adaptive e-net Lasso, which only adapts the tuning parameters of the L_1 norm, DAELasso allows for tuning parameters of both the L_1 and L_2 norms to be adapted.[1] While the Lasso generally only picks one variable from among a group of highly correlated variables, DAELasso has the potential to select all of the important variables, by encouraging grouping effects through e-net and adaptive shrinkage. Our scale mixture prior leads to closed-form conditional posteriors that can be drawn from a Gibbs sampler directly. DAELasso is more flexible than its frequentist counterparts, as it does not need to impose unrealistic restrictions on the covariance matrix of the VAR

[1] Note that if we have both formal and informal economic theory at hand with which to group the data, it can be more desirable to have a different type of Lasso, such as the grouped Lasso for VAR shrinkage, instead of the e-net Lasso. However, we do not generally have such information, and the e-net Lasso turns out to be the most appealing choice, as it encourages the grouping effect which is supported by the data itself (Zou & Hastie, 2005).

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errors. Hence, it can capture the interdependencies between the variables better. Considering that DAELasso can be too complicated for some data, we also introduce four alternative Lasso types of VAR shrinkage methods in this paper: Lasso, adaptive Lasso, e-net Lasso and adaptive e-net Lasso, each of which is a nested version of DAELasso.

Large Bayesian VARs are used widely for forecasting macroeconomic variables (e.g., Banbura, Giannone, & Reichlin, 2010; Sims, 1972, 1980). In empirical work, we evaluate the forecasting performances of the DAELasso approach and its variants. In addition, we compare the forecasting performances of these Lasso types of methods with those of the popular Baysian VAR shrinkage methods reviewed by Koop (2013). Those priors include the traditional Minnesota prior of Doan, Litterman, and Sims (1984) and Litterman (1986), and its natural variants (e.g. Banbura et al., 2010; Kadiyala & Karlsson, 1997), the stochastic search variable selection (SSVS) prior of George, Sun, and Ni (2008), and the family of SSVS plus Minnesota priors of Koop (2013). We employ Koop's (2013) data set, containing 20 US macroeconomic series, which was originally compiled by James H. Stock and Mark W. Watson. The data run from 1959Q1 to 2008Q4. In line with Koop (2013), we conduct rolling and recursive forecast exercises, and calculate both the mean squared forecast error (MSFE) and predictive likelihood measures. Using relatively uninformative priors, we find that the DAELasso approach leads to forecasting results that are better than or just as good as those of other Bayesian VAR shrinkage methods. This suggests that the DAELasso approach is an appropriate complement to the available Bayesian VAR toolkit.

This paper is closely related to the growing body of literature on forecasting with many macroeconomic variables.[2] Popular methods in the existing literature include factor models (e.g., Forni, Hallin, Lippi, & Reichlin, 2000; Stock & Watson, 2002a,b) and Bayesian shrinkage, involving a wide range of prior choices (e.g. Banbura et al., 2010; De Mol, Giannone, & Reichlin, 2008). De Mol et al. (2008) and Koop (2013) find that these forecasts tend to be highly correlated, and Bayesian VARs do tend to forecast better than factor models. Our empirical results provide further evidence for the robustness of Bayesian shrinkage methods for forecasting with a large number of predictors. In addition, our findings suggest that, as was stressed by Koop (2013), the simple and less computationally costly Minnesota priors can be more attractive to practitioners, as they forecast as well as or better than their much more complicated counterparts.

The remainder of the paper is organized as follows. Section 2 develops the Bayesian DAELasso methods, and Section 3 presents four alternative Lasso types of VAR shrinkage methods that are nested in DAELasso. Section 4 presents the empirical findings, and Section 5 concludes. Finally the data list and the results for the prior sensitivity analysis are provided in the Appendix.

[2] I thank a referee for pointing this out.

2. The estimator and Bayesian methods

Without loss of generality, we assume that all of the variables are centered. Let Y be a $T \times N$ dependent variable matrix, X be a $T \times Nk$ matrix containing the k lags of each dependent variable, and B be the coefficient matrix of dimension $Nk \times N$. In matrix form, an unrestricted VAR model of N variables takes the following form:

$$Y = XB + E, \tag{1}$$

where E is a $T \times N$ matrix of *i.i.d.* error terms, with its tth row distributed as $N(0, \Sigma)$.

Given the assumptions of the error term, the likelihood function of the model in Eq. (1) can be expressed as

$$L(b, \Sigma) \propto |\Sigma|^{-\frac{T}{2}} \exp\left\{-\frac{1}{2}\mathrm{tr}(Y - XB)' \times (Y - XB)\Sigma^{-1}\right\}. \tag{2}$$

Note that when $X'X$ is not of full rank, as is often the case when we have a larger number of parameters than of observations, the least squares estimator $B_{LS} = (X'X)^{+}X'Y$ is noisy, where $(X'X)^{+}$ is the Moore–Penrose generalized inverse of $X'X$.

Vectorizing the matrices, we can transform model (1) into

$$y = (I_n \otimes X)\beta + e, \tag{3}$$

where $y = \mathrm{vec}(Y)$, $\beta = \mathrm{vec}(B)$, $e = \mathrm{vec}(E)$ and $e \sim N(0, \Sigma \otimes I_T)$. The dimension of β is $N^2k \times 1$.

We define the DAELasso estimator for a VAR as follows:

$$\hat{\beta}_{dL} = \arg\min_{\beta}\left\{[y - (I_n \otimes X)\beta]'[y - (I_n \otimes X)\beta] + \sum_{j=1}^{N^2k}\lambda_{1,j}|\beta_j| + \sum_{j=1}^{N^2k}\lambda_{2,j}\beta_j^2\right\}, \tag{4}$$

where $\lambda_{1,j}$ and $\lambda_{2,j}$, for $j = 1, 2, \ldots, N^2k$, are positive tuning parameters associated with the L_1 and L_2 penalties, respectively. We allow for different tuning parameters for different values of β_j in order to allow for different degrees of shrinkages. For notational convenience, we define $\Lambda_1 = \mathrm{diag}(\lambda_{1,1}, \lambda_{1,2}, \ldots, \lambda_{1,N^2k})$ and $\Lambda_2 = \mathrm{diag}(\lambda_{2,1}, \lambda_{2,2}, \ldots, \lambda_{2,N^2k})$. Note that DAELasso, defined in Eq. (4), is closely related to Zou and Zhang's (2009) adaptive e-net Lasso for single equations.

2.1. Priors

In the Bayesian framework, the univariate Bayesian Laplace prior, which can be expressed as a scale mixtures of normals with an exponential density (Andrews & Mallows, 1974), is used widely to enforce the sparsity induced by the L_1 penalty in the Lasso (e.g., Korobilis, 2011; Leng et al., 2010; Park & Casella, 2008). It is natural to consider an extension of the univariate Bayesian Laplace prior into multivariate analysis. However, this is not so straightforward. As was noted by van Gerven, Cseke, de Lange, and Heskes (2010) and van Gerven, Cseke, Oostenveld, and Heskes

(2009), the commonly used multivariate Laplace distributions (e.g., Eltoft, Kim, & Lee, 2006; Kotz, Kozubowski, & Podgórski, 2001) generally do not factorize into a product of univariate Laplace distributions that can be associated with the individual coefficients.

Our approach is motivated directly by van Gerven et al.'s (2009, 2010) multivariate Laplace prior for single equation models. van Gerven et al. (2010, 2009) use a scale mixture of normals to reflect their prior knowledge of the interactions between the coefficients. Our scale mixture prior is similar to theirs; however, our prior is more about ensuring that the priors associated with the L_1 norm are conditional on the unrestricted covariance matrix of the VAR errors. Conditioning on the covariance matrix of the VAR errors is important, as the posterior may not be unimodal otherwise (Park & Casella, 2008). The posterior of van Gerven et al. (2010, 2009) is not in a tractable form, and they use approximate inference methods for posterior computations. In contrast, our prior can lead to tractable conditional posteriors that can be drawn from the Gibbs sampler directly.

We consider a conditional multivariate mixture prior of the following form:

$$\begin{aligned}\pi(\beta|\Sigma,\Gamma,\Lambda_1,\Lambda_2) &\propto \prod_{j=1}^{N^2k}\left\{\frac{\sqrt{\lambda_{2,j}}}{\sqrt{2\pi}}\exp\left(-\frac{\lambda_{2,j}}{2}\beta_j^2\right)\right.\\ &\quad\times\left.\int_0^\infty \frac{1}{\sqrt{2\pi f_j(\Gamma)}}\exp\left[-\frac{1}{2f_j(\Gamma)}\beta_j^2\right]d(f_j(\Gamma))\right\}\\ &\quad\times\left\{|M|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}\Gamma' M^{-1}\Gamma\right)\right\}^2, \end{aligned}\tag{5}$$

where $\Gamma=[\gamma_1,\gamma_2,\ldots,\gamma_{N^2k}]'$, $M=\Sigma\otimes I_{Nk}$, and $f_j(\Gamma)$ is a function of Γ and Λ_1 to be defined later. In this mixture prior, the terms associated with the L_1 penalty are conditional on Σ through $f_j(\Gamma)$. This is important, as the posterior will be not unimodal otherwise, due to the 'sharp corners' of the L_1 penalty (Park & Casella, 2008). In Eq. (5), the variances of β_a and β_b for $a\neq b$ are related through M. However, β_a and β_b themselves are independent of each other.

We need to find an appropriate $f_j(\Gamma)$ which provides us with tractable posteriors. The last term in Eq. (5) takes the form of a multivariate normal distribution $\Gamma\sim N(0,M)$. For ease of exposition, we first write the $N^2k\times N^2k$ covariance matrix M as

$$M=\begin{pmatrix} M_{1,1} & \cdots & M_{1,j} & M_{1,j+1} & \cdots & M_{1,N^2k}\\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots\\ M_{j,1} & \cdots & M_{j,j} & M_{j,j+1} & \cdots & M_{j,N^2k}\\ M_{j+1,1} & \cdots & M_{j+1,j} & M_{j+1,j+1} & \cdots & M_{j+1,N^2k}\\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots\\ M_{N^2k,1} & \cdots & M_{N^2k,j} & M_{N^2k,j+1} & \cdots & M_{N^2k,N^2k}\end{pmatrix}.\tag{6}$$

Then let

$$H_j=(M_{j,j+1},\ldots,M_{j,N^2k})\begin{pmatrix} M_{j+1,j+1} & \cdots & M_{j+1,N^2k}\\ \cdots & \cdots & \cdots\\ M_{N^2k,j+1} & \cdots & M_{N^2k,N^2k}\end{pmatrix}^{-1}.$$

Next, we construct independent variables τ_j for $j=1,2,\ldots,N^2k$ using standard textbook techniques (e.g., Anderson, 2003; Muirhead, 1982).

$$\tau_1=\gamma_1+H_1(\gamma_2,\gamma_3,\ldots,\gamma_{N^2k})'\tag{7}$$

$$\tau_2=\gamma_2+H_2(\gamma_3,\gamma_4,\ldots,\gamma_{N^2k})'\tag{8}$$

$$\cdots$$

$$\tau_{N^2K-1}=\gamma_{N^2k-1}+H_{N^2k-1}\gamma_{N^2k}\tag{9}$$

$$\tau_{N^2K}=\gamma_{N^2k}.\tag{10}$$

The joint density of $\tau_1,\tau_2,\ldots,\tau_{N^2k}$ is

$$N(\tau_1|0,\sigma^2_{\gamma_1})N(\tau_2|0,\sigma^2_{\gamma_2}),\ldots,N(\tau_{N^2k}|0,\sigma^2_{\gamma_{N^2k}}),\tag{11}$$

where $\sigma^2_{\gamma_j}=M_{j,j}-H_j(M_{j,j+1},\ldots,M_{j,N^2k})'$, with $\sigma^2_{\gamma_{N^2k}}=M_{N^2k,N^2k}$. Note that it is computationally feasible to derive $\sigma^2_{\gamma_j}$ when M is sparse.

The Jacobian of transforming $\Gamma\sim N(0,M)$ to Eq. (11) is 1. Defining $\eta_j=\tau_j/\lambda_{1,j}$, we can write Eq. (11) as

$$\begin{aligned}&N(\eta_1|0,\sigma^2_{\gamma_1}\lambda^{-2}_{1,1})N(\eta_2|0,\sigma^2_{\gamma_2}\lambda^{-2}_{1,2}),\\ &\quad\ldots,N(\eta_{N^2k}|0,\sigma^2_{\gamma_{N^2k}}\lambda^{-2}_{1,N^2k}).\end{aligned}\tag{12}$$

Let $f_j(\Gamma)=2(\eta_j^2)$. Our scale mixture prior in Eq. (5) can be rewritten as:

$$\begin{aligned}\pi(\beta|\Sigma,\Gamma,\Lambda_1,\Lambda_2) &\propto \prod_{j=1}^{N^2k}\left\{\frac{\sqrt{\lambda_{2,j}}}{\sqrt{2\pi}}\exp\left(-\frac{\lambda_{2,j}}{2}\beta_j^2\right)\right.\\ &\quad\times\left.\int_0^\infty \frac{1}{\sqrt{2\pi((2\eta_j^2))}}\exp\left[-\frac{\beta_j^2}{2(2\eta_j^2)}\right]d(2\eta_j^2)\right.\\ &\quad\times\left.\frac{\lambda^2_{1,j}}{2\sigma^2_{\gamma_j}}\exp\left[-\frac{1}{2}\frac{2\eta_j^2}{(\sigma^2_{\gamma_j})/\lambda^2_{1,j}}\right]\right\}.\end{aligned}\tag{13}$$

The last two terms in Eq. (13) constitute a scale mixture of normals (with an exponential mixing density), which can be expressed as the univariate Laplace distribution $\frac{\lambda_{1,j}}{2\sqrt{\sigma^2_{\gamma_j}}}\exp\left(-\frac{\lambda_{1,j}}{\sqrt{\sigma^2_{\gamma_j}}}|\beta_j|\right)$. Readers who are familiar with the Bayesian Lasso literature can easily show that this is an adaptive version of the Laplace prior of Park and Casella (2008) that enforces the $L1$ penalty. Overall, the priors in Eq. (13) resemble the conditional prior for the e-net Lasso used by Kyung et al. (2010). However, our prior is more complicated than that of Kyung et al. (2010). First, we allow different levels of shrinkages for each of the coefficients.[3] Second, the prior distributions of the coefficients are conditional on the covariance matrix of the multivariate model.

[3] In the literature, Griffin and Brown (2011) explore the use of different priors for achieving the adaptive Lasso which adapt automatically. Leng et al. (2010) propose using varying degrees of shrinkage for the tuning parameter. Our adaptive approach is in spirit of Leng et al. (2010).

Eq. (13) shows that the conditional prior for β_j is $N(0, \frac{2\eta_j^2}{2\lambda_{2,j}\eta_j^2+1})$, and the conditional prior for β is

$$\beta|\Gamma, \Sigma, \Lambda_1, \Lambda_2 \sim N(0, D_\Gamma^*), \tag{14}$$

where

$$D_\Gamma^* = \text{diag}\left([\frac{2\eta_1^2}{2\lambda_{2,1}\eta_1^2+1}, \frac{2\eta_2^2}{2\lambda_{2,2}\eta_2^2+1}, \ldots, \frac{2\eta_{N^2k}^2}{2\lambda_{2,N^2k}\eta_{N^2k}^2+1}]\right).$$

The tightness of the prior for each β_j depends on $\frac{2\eta_j^2}{2\lambda_{2,j}\eta_j^2+1}$. If $\frac{2\eta_j^2}{2\lambda_{2,j}\eta_j^2+1}$ is small, β_j will be shrunk towards zero. If $\frac{2\eta_j^2}{2\lambda_{2,j}\eta_j^2+1}$ is large, the prior for β_j can become quite uninformative.

Priors for $\Sigma, \lambda_{1,j}^2$ and $\lambda_{2,j}$ can be elicited following standard practice in the VAR and Lasso literature. In this paper, we set the Wishart prior for Σ^{-1} and Gamma priors for $\lambda_{1,j}^2$ and $\lambda_{2,j}$: $\Sigma^{-1} \sim W(\underline{S}^{-1}, \underline{\nu})$, $\lambda_{1,j}^2 \sim G(\underline{\mu}_{\lambda_{1,j}^2}, \underline{\nu}_{\lambda_{1,j}^2})$, $\lambda_{2,j} \sim G(\underline{\mu}_{\lambda_{2,j}}, \underline{\nu}_{\lambda_{2,j}})$.[4]

2.2. Posteriors and the Gibbs sampler

Combining the priors and likelihood, the following full conditional posteriors can be derived easily.

The full conditional posterior for β is $\beta \sim N(\overline{\beta}, \overline{V}_\beta)$, where $\overline{V}_\beta = [(I_N \otimes X)'(\Sigma^{-1} \otimes I_{Nk})(I_N \otimes X) + (D_\Gamma^*)^{-1}]^{-1}$, and $\overline{\beta} = \overline{V}_\beta[(I_N \otimes X)'(\Sigma^{-1} \otimes I_{Nk})y]$. The full conditional posterior for Σ^{-1} is $W(\overline{S}^{-1}, \overline{\nu})$, with $\overline{S}^{-1} = (Y - XB)'(Y - XB) + 2Q'Q + \underline{S}^{-1}$ and $\overline{\nu} = T + 2Nk + \underline{\nu}$, with $\text{vec}(Q) = \Gamma$. The full conditional posterior for $\lambda_{1,j}^2$ is $G(\bar{\mu}_{\lambda_{1,j}}, \bar{\nu}_{\lambda_{1,j}})$, where $\bar{\nu}_{\lambda_{1,j}} = \underline{\nu}_{\lambda_{1,j}} + 2$ and $\bar{\mu}_{\lambda_{1,j}} = \frac{\bar{\nu}_{\lambda_{1,j}}\sigma_j^2\underline{\mu}_{\lambda_{1,j}}}{2\tau_j^2\underline{\mu}_{\lambda_{1,j}} + \underline{\nu}_{\lambda_{1,j}}\sigma_{\gamma_j}^2}$. The full conditional posterior for $\lambda_{2,j}$ is $G(\bar{\mu}_{\lambda_{2,j}}, \bar{\nu}_{\lambda_{2,j}})$, where $\bar{\nu}_{\lambda_{2,j}} = \underline{\nu}_{\lambda_{2,j}} + 1$ and $\bar{\mu}_{\lambda_{2,j}} = \frac{\underline{\mu}_{\lambda_{2,j}}\bar{\nu}_{\lambda_{2,j}}}{\underline{\nu}_{\lambda_{2,j}} + \underline{\mu}_{\lambda_{2,j}}\beta_j^2}$. Finally, the full conditional posterior of $\frac{1}{2\eta_j^2}$ is inverse Gaussian: $IG(\sqrt{\frac{\lambda_{1,j}^2}{\beta_j^2\sigma_{\gamma_j}^2}}, \frac{\lambda_{1,j}^2}{\sigma_{\gamma_j}^2})$.[5] Γ cannot be drawn from the posteriors directly, but it can be recovered in each Gibbs iteration using the draws of $\frac{1}{2\eta_j^2}$ and Σ.

Conditional on arbitrary starting values, the Gibbs sampler contains the following six steps:

1. draw $\beta|\Sigma, \Lambda_1, \Lambda_2$ and Γ from $N(\overline{\beta}, \overline{V}_\beta)$;
2. draw $\Sigma^{-1}|\beta, \Lambda_1, \Lambda_2$ and Γ from $W(\overline{S}^{-1}, \overline{\nu})$;
3. draw $\lambda_{1,j}^2|\beta, \Sigma, \Lambda_{1,-j}, \Lambda_2$ and Γ from $G(\bar{\mu}_{\lambda_{1,j}}, \bar{\nu}_{\lambda_{1,j}})$ for $j = 1, 2, \ldots, N^2k$;
4. draw $\lambda_{2,j}|\beta, \Sigma, \Lambda_1, \Lambda_{2,-j}$ and Γ from $G(\bar{\mu}_{\lambda_{2,j}}, \bar{\nu}_{\lambda_{2,j}})$ for $j = 1, 2, \ldots, N^2k$;
5. draw $\frac{1}{2\eta_j^2}|\beta, \Sigma, \Lambda_1$ and Λ_2 from $IG\left(\sqrt{\frac{\lambda_{1,j}^2}{\beta_j^2\sigma_{\gamma_j}^2}}, \frac{\lambda_{1,j}^2}{\sigma_{\gamma_j}^2}\right)$ for $j = 1, 2, \ldots, N^2k$;
6. calculate Γ based on draws of Σ and $\frac{1}{2\eta_j^2}$ in the current iteration.

3. Related Lasso types of VAR shrinkage

DAELasso provides a general method for shrinking both the variable and parameter space of a VAR. However, with the number of tuning parameters being twice the number of coefficients, DAELasso might be subject to the criticism of demanding too much from the data. In this section, we introduce four alternative scaled mixture priors for β that are associated with the Lasso, adaptive Lasso, e-net Lasso, and adaptive e-net Lasso, respectively. Note that these four Lassos are all nested in DAELasso. Thus, their posteriors can be worked out easily using the procedures presented for DAELasso shrinkage. For the sake of brevity, we relegate the technical details to the online appendix (see Appendix C).

3.1. Lasso VAR shrinkage

Following Song and Bickel (2011), we define the Lasso estimator for a VAR as:

$$\hat{\beta}_L = \arg\min_\beta \left\{[y - (I_n \otimes X)\beta]'[y - (I_n \otimes X)\beta] + \lambda_1 \sum_{j=1}^{N^2k} |\beta_j|\right\}. \tag{15}$$

Correspondingly, the conditional multivariate mixture prior for β takes the following form:

$$\pi(\beta|\Sigma, \Gamma, \lambda_1) \propto \prod_{j=1}^{N^2k}\left\{\int_0^\infty \frac{1}{\sqrt{2\pi f_j(\Gamma))}} \times \exp\left[-\frac{1}{2f_j(\Gamma)}\beta_j^2\right] d(f_j(\Gamma))\right\} \times \left\{|M|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}\Gamma' M^{-1}\Gamma\right)\right\}^2. \tag{16}$$

Let $f_j(\Gamma) = 2(\eta_j^2)$; then, the scale mixture prior is:

$$\pi(\beta|\Sigma, \Gamma, \lambda_1) \propto \prod_{j=1}^{N^2k}\left\{\int_0^\infty \frac{1}{\sqrt{2\pi((2\eta_j^2))}} \times \exp\left[-\frac{\beta_j^2}{2(2\eta_j^2)}\right] d(2\eta_j^2)\right.$$

[4] Please refer to Koop (2003, p. 326) for the Gamma distribution, and Zellner (1971, p. 389) for the Wishart distribution.

[5] We adopt the form of the inverse Gaussian density used by Park and Casella (2008).

$$\times \frac{\lambda_1^2}{2\sigma_{\gamma_j}^2} \exp\left[-\frac{1}{2}\frac{2\eta_j^2}{(\sigma_{\gamma_j}^2)/\lambda_1^2}\right]\Bigg\}, \quad (17)$$

where $\eta_j = \tau_j/\lambda_1$.

3.2. Adaptive Lasso VAR shrinkage

We define the adaptive Lasso estimator for a VAR as:

$$\hat{\beta}_{\mathrm{AL}} = \arg\min_{\beta}\Bigg\{[y-(I_n\otimes X)\beta]'[y-(I_n\otimes X)\beta] + \sum_{j=1}^{N^2k}\lambda_{1,j}|\beta_j|\Bigg\}. \quad (18)$$

Correspondingly, the conditional multivariate mixture prior for β takes the following form:

$$\pi(\beta|\Sigma,\Gamma,\Lambda_1) \propto \prod_{j=1}^{N^2k}\Bigg\{\int_0^\infty \frac{1}{\sqrt{2\pi f_j(\Gamma))}} \times \exp\left[-\frac{1}{2f_j(\Gamma)}\beta_j^2\right]d(f_j(\Gamma))\Bigg\} \times \Bigg\{|M|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}\Gamma' M^{-1}\Gamma\right)\Bigg\}^2. \quad (19)$$

Let $f_j(\Gamma) = 2(\eta_j^2)$; then, the scale mixture prior is:

$$\pi(\beta|\Sigma,\Gamma,\Lambda_1) \propto \prod_{j=1}^{N^2k}\Bigg\{\int_0^\infty \frac{1}{\sqrt{2\pi((2\eta_j^2))}} \times \exp\left[-\frac{\beta_j^2}{2(2\eta_j^2)}\right]d(2\eta_j^2) \times \frac{\lambda_{1,j}^2}{2\sigma_{\gamma_j}^2}\exp\left[-\frac{1}{2}\frac{2\eta_j^2}{(\sigma_{\gamma_j}^2)/\lambda_{1,j}^2}\right]\Bigg\}, \quad (20)$$

where $\eta_j = \tau_j/\lambda_{1,j}$.

3.3. E-net Lasso VAR shrinkage

We define the e-net Lasso estimator for a VAR as:

$$\hat{\beta}_{\mathrm{EL}} = \arg\min_{\beta}\Bigg\{[y-(I_n\otimes X)\beta]'[y-(I_n\otimes X)\beta] + \lambda_1\sum_{j=1}^{N^2k}|\beta_j| + \lambda_2\sum_{j=1}^{N^2k}\beta_j^2\Bigg\}. \quad (21)$$

Correspondingly, the conditional multivariate mixture prior for β takes the following form:

$$\pi(\beta|\Sigma,\Gamma,\lambda_1,\lambda_2) \propto \prod_{j=1}^{N^2k}\Bigg\{\frac{\sqrt{\lambda_2}}{\sqrt{2\pi}}\exp\left(-\frac{\lambda_2}{2}\beta_j^2\right) \times \int_0^\infty \frac{1}{\sqrt{2\pi f_j(\Gamma))}}\exp\left[-\frac{1}{2f_j(\Gamma)}\beta_j^2\right]d(f_j(\Gamma))\Bigg\} \times \Bigg\{|M|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}\Gamma' M^{-1}\Gamma\right)\Bigg\}^2. \quad (22)$$

Let $f_j(\Gamma) = 2(\eta_j^2)$; then, the scale mixture prior is:

$$\pi(\beta|\Sigma,\Gamma,\lambda_1,\lambda_2) \propto \prod_{j=1}^{N^2k}\Bigg\{\frac{\sqrt{\lambda_2}}{\sqrt{2\pi}}\exp\left(-\frac{\lambda_2}{2}\beta_j^2\right) \times \int_0^\infty \frac{1}{\sqrt{2\pi((2\eta_j^2))}}\exp\left[-\frac{\beta_j^2}{2(2\eta_j^2)}\right]d(2\eta_j^2) \times \frac{\lambda_1^2}{2\sigma_{\gamma_j}^2}\exp\left[-\frac{1}{2}\frac{2\eta_j^2}{(\sigma_{\gamma_j}^2)/\lambda_1^2}\right]\Bigg\}, \quad (23)$$

where $\eta_j = \tau_j/\lambda_1$.

3.4. Adaptive e-net Lasso VAR shrinkage

In line with Zou and Zhang (2009), we define the adaptive e-net Lasso estimator for a VAR as

$$\hat{\beta}_{\mathrm{AEL}} = \arg\min_{\beta}\Bigg\{[y-(I_n\otimes X)\beta]'[y-(I_n\otimes X)\beta] + \sum_{j=1}^{N^2k}\lambda_{1,j}|\beta_j| + \lambda_2\sum_{j=1}^{N^2k}\beta_j^2\Bigg\}. \quad (24)$$

Correspondingly, the conditional multivariate mixture prior for β takes the following form:

$$\pi(\beta|\Sigma,\Gamma,\Lambda_1,\lambda_2) \propto \prod_{j=1}^{N^2k}\Bigg\{\frac{\sqrt{\lambda_2}}{\sqrt{2\pi}}\exp\left(-\frac{\lambda_2}{2}\beta_j^2\right) \times \int_0^\infty \frac{1}{\sqrt{2\pi f_j(\Gamma))}}\exp\left[-\frac{1}{2f_j(\Gamma)}\beta_j^2\right]d(f_j(\Gamma))\Bigg\} \times \Bigg\{|M|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}\Gamma' M^{-1}\Gamma\right)\Bigg\}^2. \quad (25)$$

Let $f_j(\Gamma) = 2(\eta_j^2)$. The scale mixture prior in Eq. (25) can be rewritten as:

$$\pi(\beta|\Sigma,\Gamma,\Lambda_1,\lambda_2) \propto \prod_{j=1}^{N^2k}\Bigg\{\frac{\sqrt{\lambda_2}}{\sqrt{2\pi}}\exp\left(-\frac{\lambda_2}{2}\beta_j^2\right) \times \int_0^\infty \frac{1}{\sqrt{2\pi((2\eta_j^2))}}\exp\left[-\frac{\beta_j^2}{2(2\eta_j^2)}\right]d(2\eta_j^2) \times \frac{\lambda_{1,j}^2}{2\sigma_{\gamma_j}^2}\exp\left[-\frac{1}{2}\frac{2\eta_j^2}{(\sigma_{\gamma_j}^2)/\lambda_{1,j}^2}\right]\Bigg\}, \quad (26)$$

where $\eta_j = \tau_j/\lambda_{1,j}$.

4. Empirical illustration

4.1. Data

In macroeconomics, it is a standard practice to assess models based on their forecasting performances (e.g., Giannone, Lenza, & Primiceri, 2010; Litterman, 1986). Koop (2013) provides an extensive forecast evaluation for seven popular Bayesian VAR priors. We employ the data set of Koop (2013), an updated version of that used by Stock and Watson (2008), for the out-of-sample forecasting analysis. The data set contains twenty quarterly macroeconomic series, including a measure of economic activity (GDP, real GDP), prices (CPI, the consumer price index), an interest rate (FFR, the Fed funds rate), and seventeen other variables.[6] Four of the seventeen variables are those used in the monetary model of Christiano, Eichenbaum, and Evans (1999). The other thirteen variables contain important aggregated information on the economy. The time series span the period from 1959Q1 to 2008Q4. A full list of the variables is provided in Appendix A. For a detailed description of the data, please refer to Koop (2013) and Stock and Watson (2008). The data are transformed to stationarity and standardized, as was done by Koop (2013).[7]

4.2. Forecast evaluation

As was done by Koop (2013), we conduct rolling and recursive forecast exercises, and calculate both the mean squared forecast error (MSFE) and predictive likelihood measures using a reduced form VAR of order four. The window length for the rolling estimation is set to be ten years. Both the recursive and rolling forecasts are conducted for $t_0 + h, t_0 + 1 + h, \ldots, T$, where t_0 is 1969Q4. Let y^f_{t+h} be the hth period forecast of y using data available at time t, and let y_{t+h} be the real value for y observed at $t + h$. The MSFE for the variable y_i is calculated as an average of the mean squared errors of the point estimates:

$$\text{MSFE} = \frac{\sum_{t=t_0}^{T-h} [y_{i,t+h} - E(y^f_{i,t+h} \mid \text{Data}_t)]^2}{T - h - t_0 + 1}. \tag{27}$$

The predictive likelihood is used to evaluate the entire predictive distribution. In particular, the following sum of the log predictive likelihood is used:

$$\sum_{t=t_0}^{T-h} \log[p(y^f_{i,t+h} = y_{i,t+h} \mid \text{Data}_t)]. \tag{28}$$

[6] These 20 variables are used by Koop (2013) for a medium-sized VAR. Koop (2013) also examines the VAR forecasts using a medium-large VAR, which contains 40 variables, and a large VAR, which contains 168 variables. In this paper, we only consider Koop's (2013) medium-size VAR, for two reasons. First, it is computationally costly to use DAELasso priors to estimate the medium-large and large VARs. Second, it has been shown in the literature (e.g., Banbura et al., 2010, Koop, 2013) that most of the gains in forecasting performance are achieved by using medium VARs with about 20 key variables.

[7] I am grateful to Mark Watson for providing the data. In addition, I am grateful to Gary Koop for sharing the Matlab code for the data transformation.

Table 1
Rolling forecasting for $h = 1$.

	GDP	CPI	FFR
DAELasso	0.58 (−198.9)	0.32 (−192.7)	0.57 (−211.7)
Adaptive e-net Lasso	0.67 (−195.8)	0.40 (−199.4)	0.63 (−215.0)
Adaptive Lasso	0.77 (−225.6)	0.31 (−209.2)	0.62 (−228.3)
Minn. Prior as in BGR	0.58 (−190.5)	0.34 (−209.2)	0.51 (−177.4)
SSVS non-conj. plus Minn. Prior	0.68 (−197.9)	0.34 (−195.2)	0.52 (−177.2)

Notes: The values are MSFEs as a proportion of the random walk MSFEs. The sums of log predictive likelihoods are given in parentheses.

For DAELasso, we need to elicit priors for $\lambda^2_{1,j}$, $\lambda_{2,j}$, and Σ. It is practically impossible to set informative priors for each $\lambda^2_{1,j}$ and $\lambda_{2,j}$, and thus we set relatively uninformative priors for $\lambda^2_{1,j}$ (or λ^2_1) and $\lambda_{2,j}$ (or λ_2) to be $G(1, 0.001)$ and $G(1, 0.01)$, respectively. The prior for Σ^{-1} is set to be $W((N - 1)I_N, 1)$, which is also relatively uninformative. There is room for improving the forecasting performance of DAELasso, for example by eliciting more informative priors. We do not explore this possibility in the current paper, because the goal of our exercise is to determine whether a DAELasso with relatively uninformative priors can provide acceptable forecasting results. For purposes of comparison, the priors for the Lasso, adaptive Lasso, e-net Lasso, and adaptive e-net Lasso are all set in the same manner. Following Geweke and Amisano (2011) and Koop (2013), we calculate the log predictive likelihood in each replication of the Gibbs sampler using posterior draws of the parameters, then take the average when the Gibbs ends. Again for comparison, we also use the mean posterior parameters to calculate the log predictive likelihood. It turns out that the results are similar.

In order to save space, we relegate a comprehensive comparison of the forecasts of DAELasso and its variants with those of the priors used by Koop (2013) to the online appendix (see Appendix C). In this section, we only report the forecasts results for the five most important priors: DAELasso, the adaptive e-net Lasso, the adaptive Lasso, the natural conjugate prior used by Banbura et al. (2010), which is labelled 'Minn. Prior as in BGR', and a combination of the conjugate SSVS prior and the Minnesota prior, which is labelled 'SSVS Conjugate plus Minn. Prior', and was proposed by Koop (2013). The last two priors were investigated by Koop (2013). We refer to Koop (2013) for a lucid description of these priors.

The results presented in Tables 1–4 show that the forecasting performance of the DAELasso approach is comparable to those of the popular Bayesian shrinkage methods explored by Koop (2013). Overall, the forecasts of all methods are highly correlated with each other. Compared to the priors investigated by Koop (2013), DAELasso and its variants tend to forecast better for GDP and CPI in terms of point forecasts, but not as well for FFR. The results yielded by the DAELasso approach for the predictive log-likelihood

Table 2
Rolling forecasting for $h = 4$.

	GDP	CPI	FFR
DAELasso	0.55 (−206.9)	0.48 (−205.9)	0.65 (−230.9)
Adaptive e-net Lasso	0.53 (−195.7)	0.47 (−204.4)	0.55 (−219.9)
Adaptive Lasso	0.74 (−233.9)	0.54 (−223.0)	0.78 (−247.7)
Minn. Prior as in BGR	0.59 (−217.1)	0.55 (−227.7)	0.59 (−213.4)
SSVS non-conj. plus Minn. Prior	0.63 (−209.9)	0.51 (−201.3)	0.58 (−198.1)

Notes: The values are MSFEs as a proportion of the random walk MSFEs. The sums of log predictive likelihoods are given in parentheses.

Table 3
Recursive forecasting for $h = 1$.

	GDP	CPI	FFR
DAELasso	0.55 (−210.4)	0.29 (−190.9)	0.56 (−224.2)
Adaptive e-net Lasso	0.67 (−242.0)	0.40 (−201.6)	0.63 (−239.8)
Adaptive Lasso	0.62 (−219.2)	0.28 (−196.4)	0.60 (−226.8)
Minn. Prior as in BGR	0.56 (−192.3)	0.30 (−195.9)	0.51 (−229.1)
SSVS non-conj. plus Minn. Prior	0.65 (−203.9)	0.29 (−187.6)	0.54 (−228.9)

Notes: The values are MSFEs as a proportion of the random walk MSFEs. The sums of log predictive likelihoods are given in parentheses.

Table 4
Recursive forecasting for $h = 4$.

	GDP	CPI	FFR
DAELasso	0.54 (−218.3)	0.48 (−206.6)	0.61 (−239.6)
Adaptive e-net Lasso	0.53 (−215.6)	0.47 (−207.0)	0.55 (−247.3)
Adaptive Lasso	0.63 (−228.0)	0.52 (−214.7)	0.66 (−242.2)
Minn. Prior as in BGR	0.61 (−214.7)	0.52 (−219.4)	0.59 (−249.6)
SSVS non-conj. plus Minn. Prior	0.67 (−219.0)	0.49 (−201.6)	0.53 (−233.7)

Notes: The values are MSFEs as a proportion of the random walk MSFEs. The sums of log predictive likelihoods are given in parentheses.

are more mixed. However, DAELasso's performances are qualitatively similar to those of the Bayesian VAR methods studied by Koop (2013).

When we focus on the three Lasso types of forecasts, we find that DAELasso tends to provide the best GDP and FFR point forecasts for $h = 1$, while the adaptive Lasso gives the best CPI point forecasts. These results seem to suggest that, while GDP and FFR can be forecast best by using many variables that might be highly collinear, CPI can be forecast better by using a smaller number of important variables that are not highly correlated with each other. Turning to the point forecasts for $h = 4$, we find that the adaptive e-net Lasso performs best in most cases. The forecast results for the predictive log-likelihood vary for the three Lasso types of methods, but overall, DAELasso and the adaptive e-net Lasso tend to outperform the adaptive Lasso in most cases. This result suggests the importance of using e-net to capture the possible grouping effect.

We conduct a prior sensitivity analysis using relatively tight priors. The results are reported in Appendix B. We find that the tighter priors tend to give better forecasts. However, the most important information obtained from the exercise is the fact that the general pattern found in Tables 1–4 is robust to the choice of priors.

4.3. Sparsity or stability?

As was noted by De Mol et al. (2008), in the presence of collinearity, if a shrinkage method only enforces sparsity, as the Lasso does, it will be very sensitive to minor perturbations of the data. The traditional wisdom says to use the e-net Lasso to enforce both the $L1$ and $L2$ penalties (e.g., Zou & Hastie, 2005), but it is less clear whether using the e-net Lasso is the most effective way to improve the stability of variable selections. In this paper, we investigate this issue by examining the posterior mean estimates for DAELasso and its variants. We find that, using similar priors, the e-net Lasso tends to enforce slightly less sparsity than the Lasso, though DAELasso, the adaptive e-net Lasso and the adaptive Lasso all tend to lead to huge improvements in the stability of variable selection by pulling the parameters of all of the highly collinear variables towards zero. Our findings suggest that adaptive shrinkage is more effective than e-net in terms of enforcing stability.[8]

Tables 5–7 present the correlations between the posterior mean estimates for the coefficients in the GDP, CPI, and FFR equations for DAELasso and its variants. Here, the coefficients for all of the recursive estimations are pooled together.[9] Interestingly, we find that the coefficients for the adaptive types of models are always highly correlated, and the coefficients for the e-net Lasso and the Lasso are also highly correlated. This implies that the degree of sparsity which is enforced is affected more by whether or not adaptive shrinkage is used.

Table 8 presents the means and standard deviations for the posterior estimates of the coefficients in the GDP, CPI, and FFR equations. In general, the results for the three adaptive types of Lassos are very similar, as are those for the e-net Lasso and the Lasso. A closer look reveals that the e-net Lasso yields posterior estimates that are slightly more centered at zero than those from the Lasso. However, the most prominent pattern that leaps out of Table 8 is the fact that the standard deviations for the coefficients in the e-net Lasso and Lasso are almost ten times lager than those of their counterparts in the three adaptive types of models.

[8] Our finding that adaptive types of Lasso models tend to give more stable variable selection results, however, is based on priors which are the same as or similar to those we elicited for the shrinkage parameters across models. This scenario may change if the priors for the shrinkage parameters of the different models are very different.

[9] The results for rolling estimates are very similar.

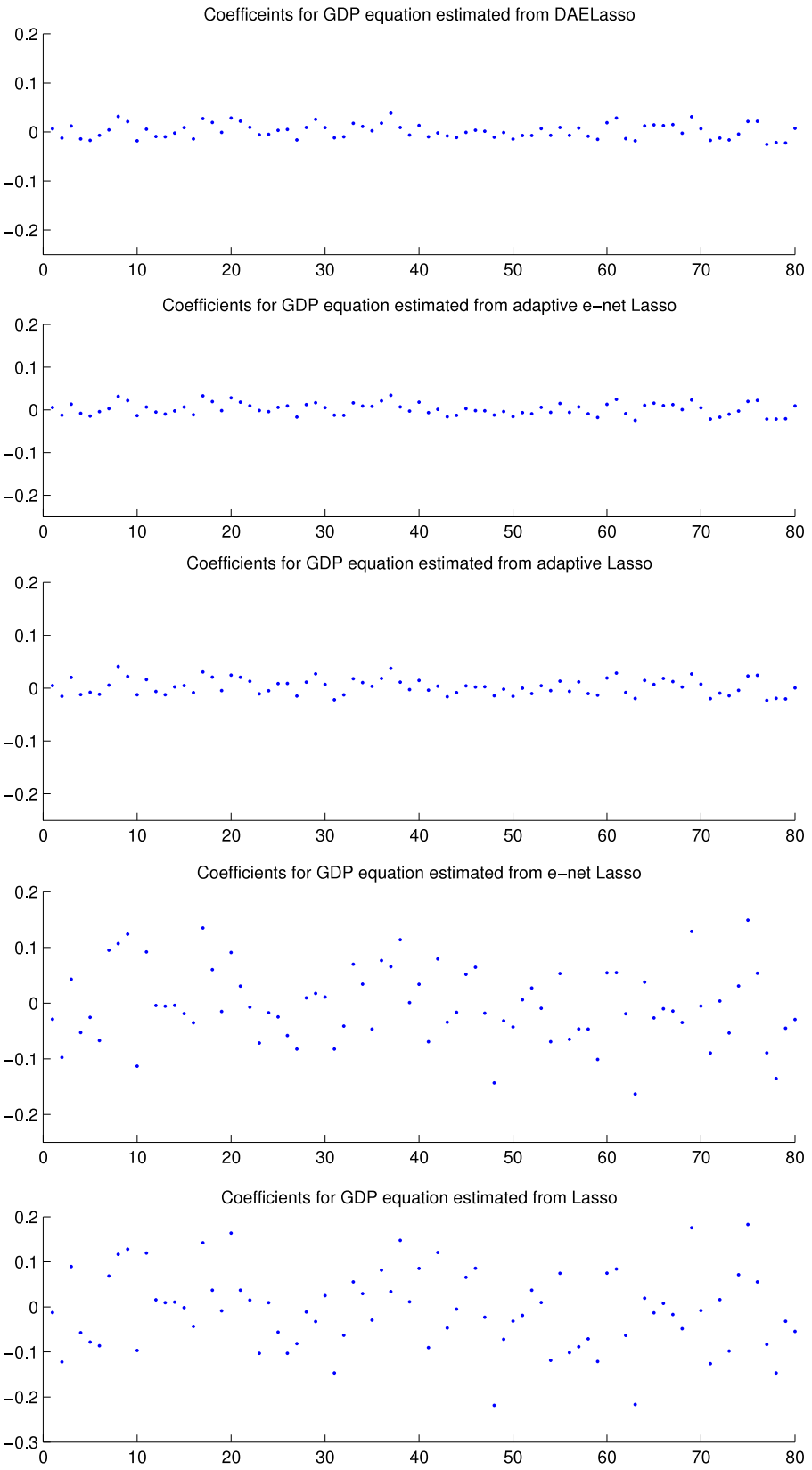


Fig. 1. The coefficients plots.

Table 5
Correlations between the coefficients in the GDP equation.

	DAELasso	Adaptive e-net Lasso	Adaptive Lasso	e-net Lasso	Lasso
DAELasso	1.00				
Adaptive e-net Lasso	0.97	1.00			
Adaptive Lasso	0.97	0.97	1.00		
e-net Lasso	0.70	0.70	0.69	1.00	
Lasso	0.66	0.65	0.66	0.97	1.00

Table 6
Correlations between the coefficients in the CPI equation.

	DAELasso	Adaptive e-net Lasso	Adaptive Lasso	e-net Lasso	Lasso
DAELasso	1.00				
Adaptive e-net Lasso	0.97	1.00			
Adaptive Lasso	0.97	0.97	1.00		
e-net Lasso	0.73	0.74	0.73	1.00	
Lasso	0.69	0.69	0.69	0.96	1.00

Table 7
Correlations between the coefficients in the FFR equation.

	DAELasso	Adaptive e-net Lasso	Adaptive Lasso	e-net Lasso	Lasso
DAELasso	1.00				
Adaptive e-net Lasso	0.97	1.00			
Adaptive Lasso	0.98	0.97	1.00		
e-net Lasso	0.74	0.76	0.74	1.00	
Lasso	0.71	0.71	0.71	0.97	1.00

Table 8
Means and standard deviations for the coefficients.

		DAELasso	Adaptive e-net Lasso	Adaptive Lasso	e-net Lasso	Lasso
GDP	Mean	0.0021	0.0020	0.0021	−0.0013	−0.0029
	Std.	(0.0191)	(0.0180)	(0.0191)	(0.1032)	(0.1267)
CPI	Mean	0.0013	0.0012	0.0013	0.0000	−0.0013
	Std.	(0.0193)	(0.0184)	(0.0193)	(0.0799)	(0.0943)
FFR	Mean	0.0053	0.0053	0.0054	0.0130	0.0150
	Std.	(0.0222)	(0.0207)	(0.0222)	(0.0939)	(0.1062)

As an example, Fig. 1 plots the posterior estimates of the coefficients in the GDP equation for the whole sample. The coefficients are plotted against the sequence number of the variables. The similarities between the coefficient plots for DAELasso, the adaptive e-net Lasso, and the adaptive Lasso are striking, as are the similarities between the coefficient plots for the e-net Lasso and the Lasso. As the coefficients estimated using the adaptable types of Lassos are much closer to zero than their counterparts estimated using the e-net Lasso and the Lasso, it is understandable that minor perturbations in the data tend to have greater impacts on the e-net Lasso and Lasso forecasts. For the sake of brevity, we do not present comparable figures for other variables and sample periods, but they all support our general findings.

5. Conclusion

This paper proposes a Bayesian DAELasso approach to VAR shrinkage. We elicit a scale mixture prior which leads to closed-form conditional posteriors that can be drawn from a Gibbs sampler directly. The method is appealing, as it can achieve variable selection and coefficient shrinkage simultaneously in a data based fashion. DAELasso deals with the multicollinearity problem constructively by encouraging the grouping effect through both the e-net and adaptive shrinkages. Hence, its forecasting results can be more stable and less subject to the influence of minor data perturbations. Using a relatively uninformative prior, we find that the forecasting performance of DAELasso is comparable to that of other popular Bayesian VAR shrinkage methods. This shows that the DAELasso approach can be used as an appropriate addition to the available Bayesian VAR toolkits. The implementation of the DAELasso approach is simple and straightforward. It can easily be extended to a nonlinear framework in order to shed new light on macroeconomic analysis and forecasting.

It is interesting to find that the sophisticated and well-designed priors we use for DAELasso and its variants produce forecasts that are correlated with and as accurate as those obtained using either a simple Minnesota prior or the conjugate priors which are traditionally used to conduct inference in large and medium-sized Bayesian VARs. This result has important implications in a broader context.

Table 9
The list of variables.

Short name	Trans. code	Data description
RGDP	5	Real GDP, quantity index (2000 = 100)
CPI	6	CPI all items
FFR	2	Interest rate: federal funds (effective) (% per annum)
Com: spot price (real)	5	Real spot market price index: all commodities
Reserves nonbor	3	Depository inst reserves: nonborrowed (mil$)
Reserves total	6	Depository inst reserves: total (mil$)
M2	6	Money stock: M2 (bil$)
Cons	5	Real personal cons. exp., quantity index
IP: total	5	Industrial production index: total
Capacity util	1	Capacity utilization: manufacturing (SIC)
U: all	2	Unemp. rate: all workers, 16 and over (%)
HStarts: total	4	Housing starts: total (thousands)
PPI: fin goods	6	Producer price index: finished goods
PCED	6	Personal consumption exp.: price index
Real AHE: goods	5	Real avg hrly earnings, non-farm prod. worker
M1	6	Money stock: M1 (bil$)
S&P: indust	5	S&Ps common stock price index: industrials
10 yr T-bond	2	Interest rate: US treasury const. mat., 10-yr
Ex rate: avg	5	US effective exchange rate: index number
Emp: total	5	Employees, nonfarm: total private

Table 10
Rolling forecasting for $h = 1$.

	GDP	CPI	FFR
DAELasso	0.58 (−189.9)	0.33 (−185.1)	0.55 (−211.2)
Adaptive e-net Lasso	0.58 (−189.9)	0.33 (−186.2)	0.55 (−212.6)
Adaptive Lasso	0.58 (−189.7)	0.33 (−185.4)	0.55 (−211.5)

Notes: The values are MSFEs as a proportion of the random walk MSFEs. The sums of log predictive likelihoods are given in parentheses.

Table 11
Rolling forecasting for $h = 4$.

	GDP	CPI	FFR
DAELasso	0.53 (−201.9)	0.47 (−204.8)	0.57 (−233.2)
Adaptive e-net Lasso	0.53 (−201.8)	0.47 (−204.6)	0.57 (−233.2)
Adaptive Lasso	0.53 (−201.8)	0.47 (−204.2)	0.57 (−232.8)

Notes: The values are MSFEs as a proportion of the random walk MSFEs. The sums of log predictive likelihoods are given in parentheses.

Table 12
Recursive forecasting for $h = 1$.

	GDP	CPI	FFR
DAELasso	0.58 (−193.9)	0.31 (−182.8)	0.54 (−226.1)
Adaptive e-net Lasso	0.59 (−194.5)	0.31 (−183.3)	0.55 (−226.6)
Adaptive Lasso	0.58 (−194.0)	0.31 (−182.6)	0.55 (−226.4)

Notes: The values are MSFEs as a proportion of the random walk MSFEs. The sums of log predictive likelihoods are given in parentheses.

First, it provides further evidence that the good forecasting performances of Bayesian VARs are a general feature of the data, and do not depend on specific features of the prior. Second, it confirms Koop's (2013) finding that simple Minnesota priors, such as the normal-conjugate prior used by Banbura et al. (2010), work well in medium and large VARs, which makes these simple priors more attractive than the computationally more demanding alternatives.

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Appendix A. Data list

We use 20 of the US macro series used by Koop (2013), who provided an updated version of the data from Stock and Watson (2008). The variables are transformed to stationarity and standardized as was done by Koop (2013). Below, we list the essential data descriptions and transformation codes from Koop (2013). Please refer to Koop (2013) for detailed data information.

The data transformation codes are as follows: 1: no transformation; 2: first difference; 3: second difference; 4: log; 5: first difference of logged variables; 6: second difference of logged variables (see Table 9).

Appendix B. Prior sensitivity analysis

Tables 10–13 report the forecast results for DAELasso, the adaptive e-net Lasso, and the adaptive Lasso using tighter priors. The priors for $\lambda_{1,j}^2$ (or λ_1^2) and $\lambda_{2,j}$ (or λ_2) are set to be $G(1, 0.0001)$ and $G(1, 0.001)$, respectively. The prior for Σ^{-1} is set to be $W(100I_N, 1)$.

Table 13
Recursive forecasting for $h = 4$.

	GDP	CPI	FFR
DAELasso	0.53 (−204.6)	0.47 (−208.6)	0.56 (−248.9)
Adaptive e-net Lasso	0.53 (−204.3)	0.47 (−208.2)	0.56 (−248.2)
Adaptive Lasso	0.53 (−204.4)	0.47 (−208.2)	0.56 (−248.1)

Notes: The values are MSFEs as a proportion of the random walk MSFEs. The sums of log predictive likelihoods are given in parentheses.

Appendix C. Supplementary data

Supplementary material related to this article can be found online at http://dx.doi.org/10.1016/j.ijforecast.2013.04.004.

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